

Lab Goal : This lab was designed to teach you more about base conversion and to allow you to review using Comparable.

Lab Description : Read in a number and its base. Convert that number to binary(100101001) and then sort all of the binary values. You will sort all binary number by first comparing the number of ones. The numbers with the smallest number of ones will be first and the numbers with the largest number of ones will be last. If two numbers have the same number of ones, sort the numbers first by the base10 value. If the base 10 values are the same, then sort the numbers by their original values. Print out the original number, its value in base10, and lastly, print out the numbers value in binary.

Sample Data :

```
15
23642 8
65535 10
5 10
6 10
33 4
99 12
12345 7
512 7
2 10
1 10
10 2
15 10
12 13
14 15
15 16
```

Java Base Conversion (allowed on this lab)

```
//converts 234 base 6 to base 10
int base10 = Integer.parseInt("234",6);

//converts base 10 to base 3
out.println(Integer.toString(base10, 3);
```

This loop produced the output:

```
out.printf("%-6s %-4s %-6s %s %n", "number", "base", "base10", "binary");
for(Number n : numList){
    String[] nStrRay = n.toString().split(" ");
    String number = nStrRay[0];
    int base = Integer.parseInt(nStrRay[1]);
    int base10 = Integer.parseInt(nStrRay[2]);
    String binary = nStrRay[3];
    out.printf("%-6s %-4d %-6d %s %n", number, base, base10, binary);
}
```

Sample Output :

number	base	base10	binary
1	10	1	1
10	2	2	10
2	10	2	10
5	10	5	101
6	10	6	110
14	15	19	10011
15	16	21	10101
12	13	15	1111
15	10	15	1111
33	4	15	1111
99	12	117	1110101
12345	7	3267	110011000011
512	7	254	11111110
23642	8	10146	10011110100010
65535	10	65535	1111111111111111